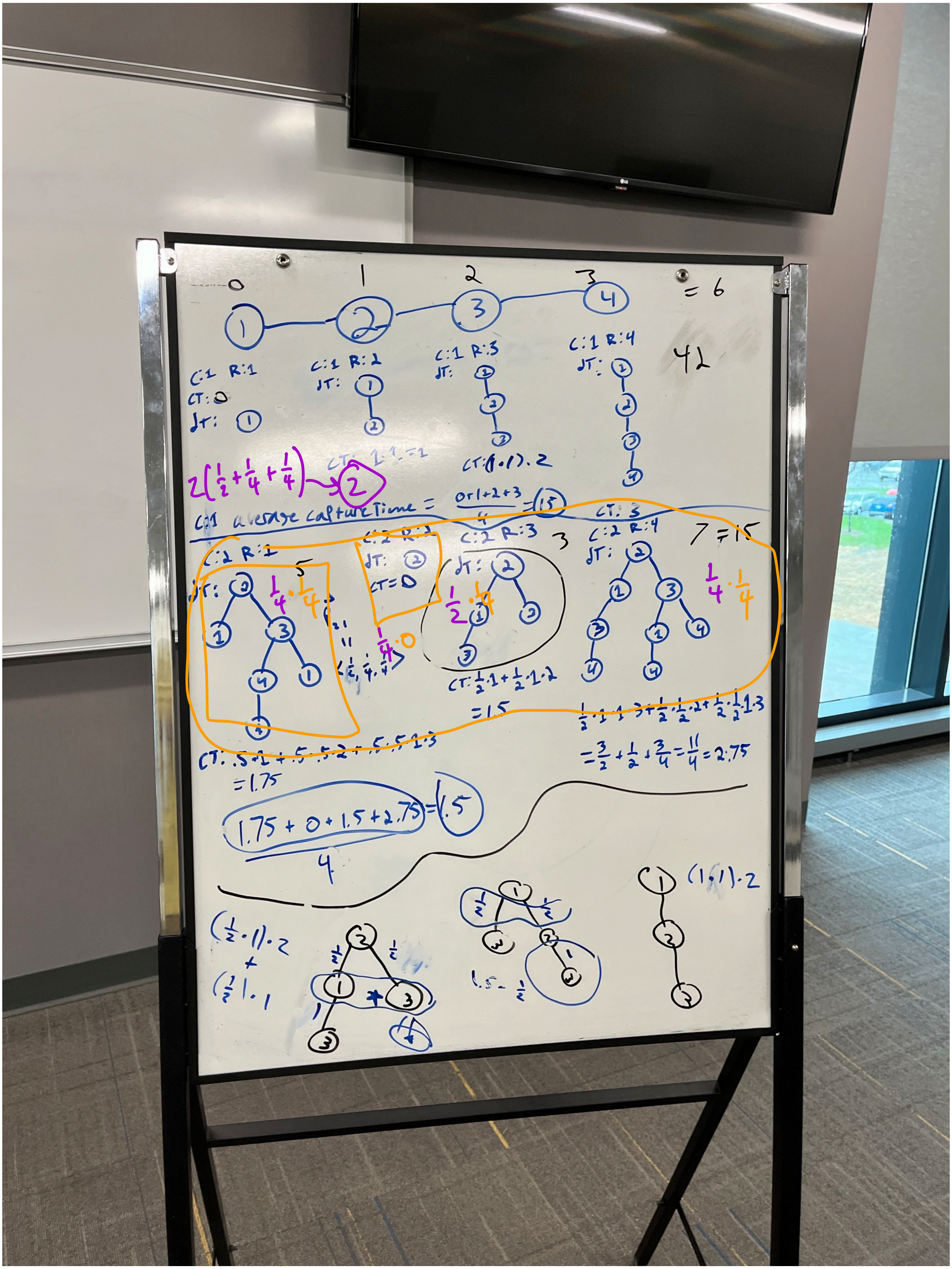
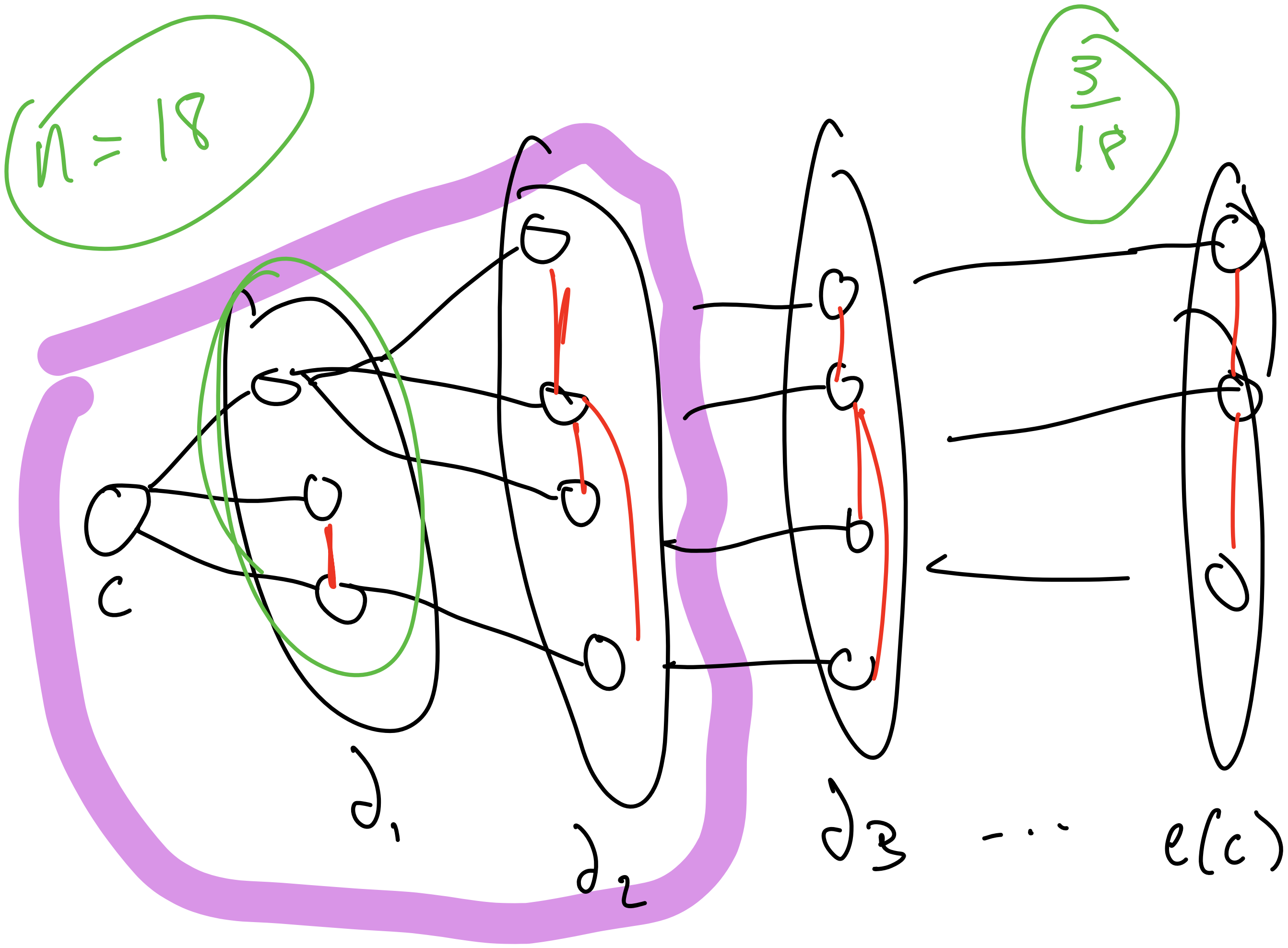


0 1 2 3
= 6
C:1 R:1
CT: 0
JT:
C:1 R:2
JT:
C:1 R:3
JT:
C:1 R:4
JT:
42
CT: (1·1)·2
2(1/2 + 1/4 + 1/4) → 2
CT average capture time = (0+1+2+3)/4 = 1.5
C:2 R:1
JT:
C:2 R:2
JT:
CT=0
C:2 R:3
JT:
C:2 R:4
JT:
CT: 3
7=15
1/4 · 1/4
1/2 · 1/4
CT: 1/2·1 + 1/2·1·2
=1.5
CT: .5·1 + .5·.5·2 + .5·.5·1·3
=1.75
= 3/2 + 1/2 + 3/4 = 11/4 = 2.75
(1.75 + 0 + 1.5 + 2.75)/4 = 1.5
(1/2·1)·2
+
(1/2 1·1
(1·1)·2
1.5 - 1/2



☆ If the sample space is the set of all possible treasure hunts H on G with cop starting on a given vertex v,

$$pr(cap(H) = k) = \frac{1}{n}$$

why 1?

for all $0 \le k \le e(v)$.

Rewording: Let H be a random hunt on G where cop starts on $v \in V(G)$. Suppose $k \in \mathbb{Z}$ with $0 \le k \le e(v)$. Then,

$$\Pr(\text{capt}(H) = k) = \frac{1}{n}.$$

★ **Corollary**: The expected "expected capture time when cop starts on u",

$$EC_{Avg}(G,u) = \left(\frac{\sum_{v \in V(G)} EC(G,u,v)}{|G|} \right)$$

is equal to

$$\sum_{k=0}^{n-1} \Pr(\text{capt}(H)=k) = \sum_{k=0}^{n-1} \frac{1}{n} k$$

$$= \frac{1}{n}\left(\sum_{k=0}^{n-1} k\right) = \frac{(n-1)n}{2} \cdot \frac{1}{n}$$

$$= \boxed{\binom{n}{2} \cdot \frac{1}{n}}$$

Desired Results:

Thrm: Let H be a random hunt on G where cop starts on $u \in V(G)$. Suppose $k \in \mathbb{Z}$ with $0 \le k \le e(u)$. Then,

$$Pr(capt(H) = k) = \frac{1}{n}.$$

Pf: Still need to prove!

Corollary: The expected, expected capture time (given cop starts on $u \in V(G)$) is $\frac{\binom{n}{2}}{n}$.

Pf:

$$E(C_{Avg}(G,u)) = \sum_{k=0}^{n-1} k \cdot Pr(capt(H)=k) = \sum_{k=0}^{n-1} \frac{1}{n} k$$

$$= \frac{1}{n}\left(\sum_{k=0}^{n-1} k\right) = \frac{(n-1)n}{2} \cdot \frac{1}{n}$$

$$= \boxed{\binom{n}{2} \cdot \frac{1}{n}} \quad \square$$

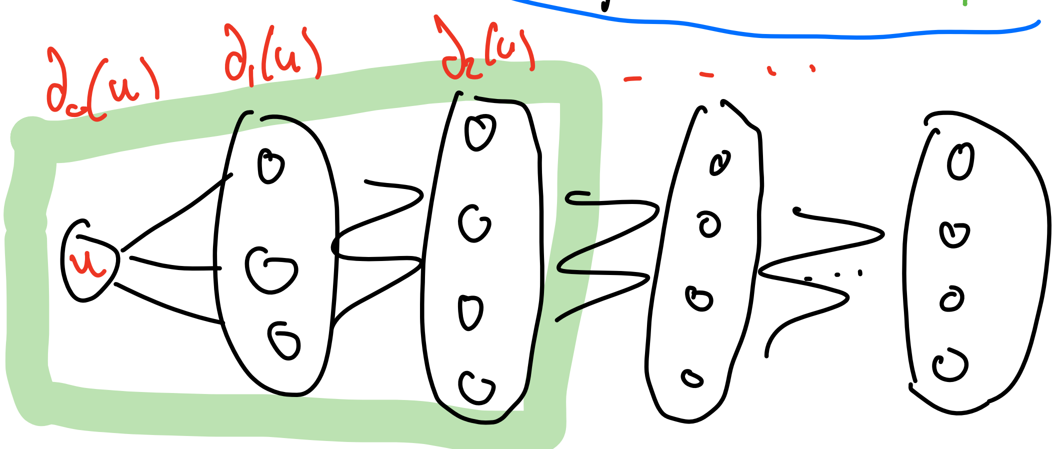
Thoughts on proving above Thrm :

- If $k=0$, there is a $\frac{1}{n}$ chance of treasure being on u (there is only one outcome where this happens)

- If $k=1$, then $pr(capt(H)=1) = \frac{\deg(u)}{n} \cdot \frac{1}{\deg(u)}$

Chance treasure is on either of those $\deg(u)$ vertices · Chance on one particular of those neighbors

- If $k=2$,



Ways to get $capt(H) = 2$:

- Treasure on $x \in \partial_2(u)$ and cop travels $u \to \partial_1(u) \to x$.

- Treasure on $x \in \partial_1(u)$ and cop moves $u \to$ "other vertex in $\partial_1(u)$" $\to x$

Other question: How big is our sample space?
↳ How many games are there in total given cop starts on u?

options for move 1

options for move 2

"if stay in ∂_1"

$$\partial_1(u) \cdot \left[(\partial_1(u) - 1) + \partial_2(u) \right] \cdot \left[\begin{array}{c} (\partial_1(u) - 1) \\ + \\ (\partial_1(u) - 2) \\ + \\ (\partial_2(u) - 1) \\ + \\ \partial_3(u) \\ + \\ \partial_2(u) \end{array} \right] \cdot \ldots$$

$$= \partial_1(u) \cdot (\partial_1(u) - 1) \cdot (\partial_1(u) - 2) \cdots (1) = [\partial_1(u)]!$$

$$= (d_0(u))! + (d_1(u))! + \dots + (d_{ecc(u)}(u))! \quad ?$$

★ Keep thinking about the computation here and why exactly $\frac{1}{n}$ hunts have a given capture time!

- What if my graph G is regular?